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**Title**

Many small carriers: the mean-field limit of competitive urban delivery under zone-time charges

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**Author contributions (CRediT)**

Quang-Vinh Dang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing - review & editing, Visualization, Project administration.

**Declaration of competing interest**

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**Data availability**

Code, instances, result files and the Lean development are available at https://github.com/vinhqdang/Competing-Carriers-under-Uncertainty (folder mfg); the city data are the processed data of the companion paper, which is under review.

**Declaration of generative AI and AI-assisted technologies in the manuscript preparation process**

During the preparation of this work the author used Claude (Anthropic) to support the writing of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

**Related manuscript**

A companion paper by the same author is under review at another journal and is cited in the manuscript as Anonymous (2026).